Modern Techniques in Modelling

LONDON SCHOOL of HYGIENE & TROPICAL MEDICINE

Who we are

Course organisers

- Katie Atkins, PhD Katherine.Atkins@lshtm.ac.uk
- Nicholas Davies, PhD Nicholas.Davies@lshtm.ac.uk
- Roz Eggo, PhD Roz.Eggo@lshtm.ac.uk
- Kaitlyn Johnson, PhD Kaitlyn.Johnson@lshtm.ac.uk

Course administration

- Francesco Grisolia Francesco.Grisolia@lshtm.ac.uk

Lecturers and Demonstrators

- Sam Abbott, Alexis Robert, Kaja Abbas (All LSHTM / CMMID-based)

Who you are (= scope of the course)

- some exposure to the theory and use of infectious disease modelling & like to start coding their own models using R

OR

- know some R but do not have experience using R to code infectious disease models

OR

- will be conducting research using infectious disease models in R

OR

- want a deeper understanding of techniques for implementing models.

Logistics

Logistics

- The days will run ~10-4.30/5pm (London) each day, with regular comfort breaks
- For online attendees:
    - Please make sure your name on Zoom is your **first name + last name or last initial** to facilitate talking to each other, and for security
    - To ask any question or raise an issue during a lecture, **please raise your hand** – a demonstrator will be monitoring the Zoom
    - Please keep cameras on when possible to encourage discussion

Resources

Resources

- All material (timetable, slides, practicals etc) on the course website:

  cmmid.github.io/mtm

  (shout out now if you haven't got access! Or cannot access wifi)

- All exercises completed using Rstudio
  (shout out now if you haven't downloaded it!)
- Lecture recordings on Moodle (ble.lshtm.ac.uk) at the end of each day
- Practical session exercises and solutions on the web site

- DISCUSSION BOARD on Moodle for **everyone**
  - please feel free to introduce yourself more fully there if you wish

What will you learn in this course?

Tuesday

Introduce our first mathematical model of infectious disease

Develop differential equation models

Wednesday

Extend differential equation models to metapopulations
Sampling, uncertainty and sensitivity analysis
Introduce some group work on a modelling problem

Thursday

Network models
Randomness and modelling

Friday

More on randomness and modelling

Group work on modelling problem

Wrap-up session

Feedback

Your feedback is important to us!

Please complete the feedback form on Moodle after the course — tell us what we did well and what we could improve.

And please don't hesitate to ask questions during the course.

Over to you!

Introduce yourself

Give your **name** + **where you work**

Use the **Moodle Discussion Board** to introduce yourself more fully should you wish or pose any questions for your colleagues